

Stochastic Strategic Demand–Capacity Balancing Using Model-Free Reinforcement Learning

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MOTIVATION

Demand–capacity balancing under weather uncertainty

Strategic demand–capacity balancing is essential to mitigating inefficiencies in air traffic operations. However, this is a challenging process, particularly due to the dynamic and uncertain nature of weather conditions.

Traditional ground-holding formulations rely on **deterministic forecasts and predefined decision schemes**, limiting their adaptability.

74.3% of U.S. system-impacting delays over 15 min are weather-related FAA, 2017–2023	20–50% of European ATFM delay is attributed to weather EUROCONTROL, 2024	40.6% of ATFM measures in Brazilian airspace due to weather DECEA, 2025
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01 Integrated RL framework

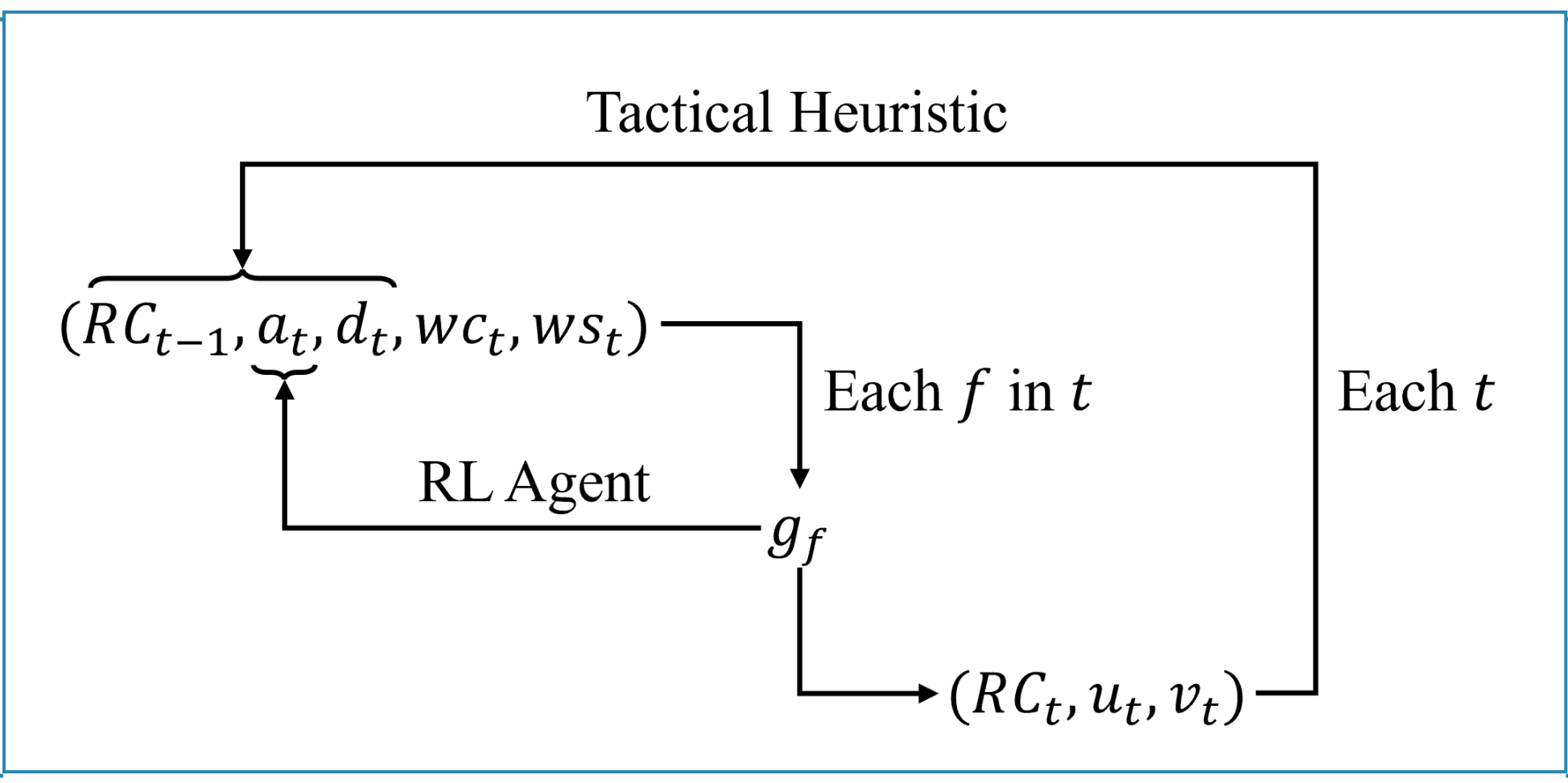
Strategic ground-delay allocation coupled with tactical capacity management under uncertain weather forecasts.

02 Two operational paradigms

The same framework applied without modification to conventional ATFM at São Paulo/Guarulhos (SBGR) and to UAM traffic management in the São Paulo Metropolitan Region.

METHOD

One loop, two decision layers



Strategic layer — RL agent

For each arrival flight f in epoch t , choose a ground delay g_f in 15-minute increments, up to one hour.

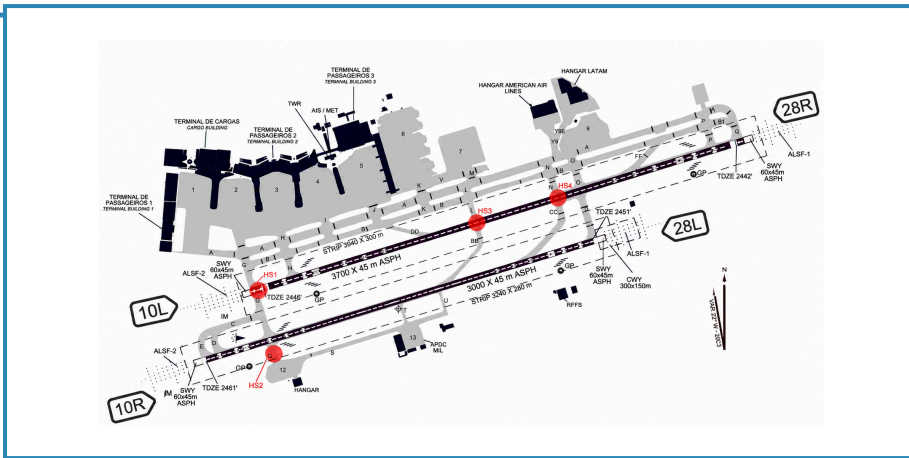
Tactical layer — heuristic

Given the wind state, select a feasible runway configuration (or vertiport side) and set arrival/departure service rates from the capacity envelope.

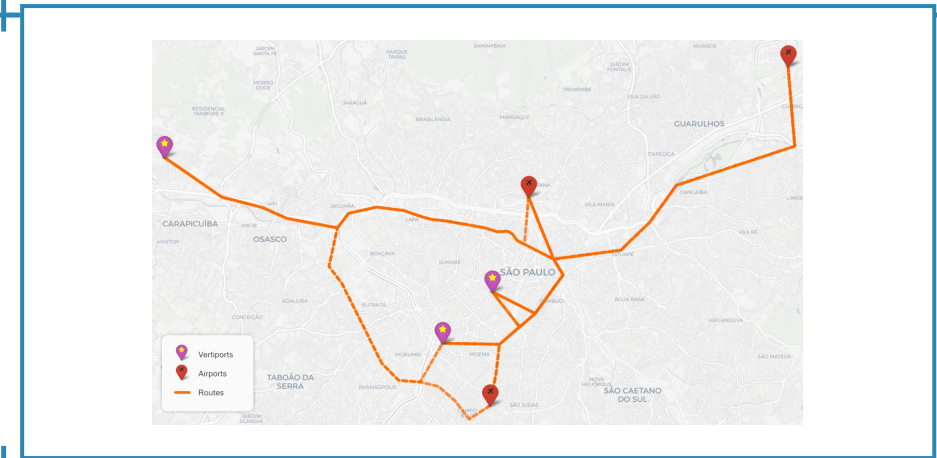
Coupling

Reallocated demand changes future queues, so the tactical consequence of each delay is fed back into the reward.

Two study settings



Case 1 — SBGR, 300k+ annual operations on two parallel runways, configurations 10R/10L and 28L/28R.



Case 2 — three vertiport candidates in the RMSP, routed to SBGR, SBSP and SBMT over the existing REH corridors.

FORMULATION

An MDP on 15-minute epochs

STATE

$$\mathbf{s}_t = (\mathbf{RC}_{t-1}, \mathbf{a}_t, \mathbf{d}_t, \mathbf{wc}_t, \mathbf{ws}_t)$$

Previous configuration · arrival queue · departure queue · weather category (VMC / IMC / LIMC) · wind state, encoding which sides stay usable at $\text{CW} \leq 20$ kt and $\text{TW} \leq 5$ kt. Horizon $T = 96$ epochs (one day).

ACTION, MASKED

$$\mathbf{g}_t \in \{0, 1, 2, 3, 4\} \times 15 \text{ min}$$

Capped at one hour and at the time remaining in the horizon; a binary mask removes infeasible delays so the agent never explores them.

TACTICAL SERVICE RATES

$$u_t = \min(u_{\max}[\mathbf{wc}_t, \mathbf{RC}_t], a_t)$$

$$v_t = \min(v_{\max}[\mathbf{wc}_t, \mathbf{RC}_t, u_t], d_t)$$

Configuration taken from the operationally valid set $\mathbf{RC}_t \in \mathcal{RC}(\mathbf{ws}_t)$.

REWARD — THE TACTICAL CONSEQUENCE

$$J^{\text{tac}} = \sum_{t=0}^T [\alpha(a_t - u_t)^2 + (d_t - v_t)^2 + \delta \cdot \mathbb{I}(\mathbf{RC}_t \neq \mathbf{RC}_{t-1}) (\alpha a_t^2 + d_t^2)]$$

$$R_t = -(\Delta J^{\text{tac}}(t) + \eta \sum_t g_t), \quad \eta = 0.1$$

A delay may relieve congestion in one epoch while worsening another, so the reward carries the full change in tactical cost over the remaining episode. Costs calibrated from EUROCONTROL (2024): €45 per delayed departure minute, €90 per airborne arrival minute.

WEATHER UNCERTAINTY

Two sources of stochasticity

1. WEATHER DYNAMICS

$$P(\text{METAR} = j \mid \text{METAR}_{t-1} = i)$$

Empirical transition frequencies between VMC, IMC and LIMC at 15-min resolution.

2. FORECAST ERROR

$$P(\text{METAR} = j \mid \text{TAF} = i)$$

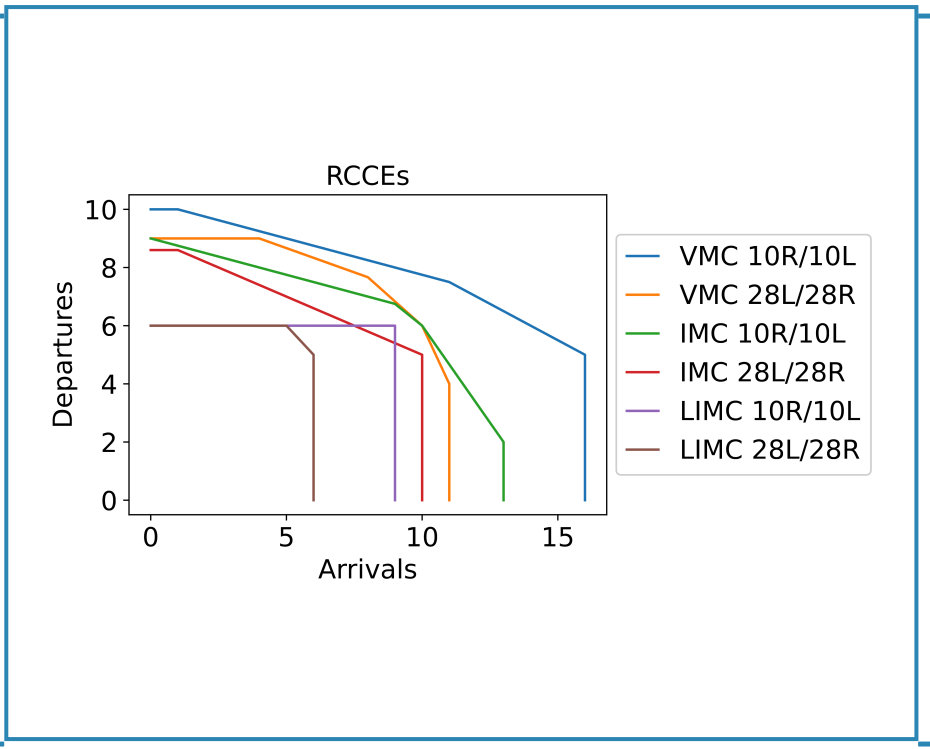
Confusion matrices estimated by comparing forecasts against observed realizations.

TRAINING-TIME SAMPLING

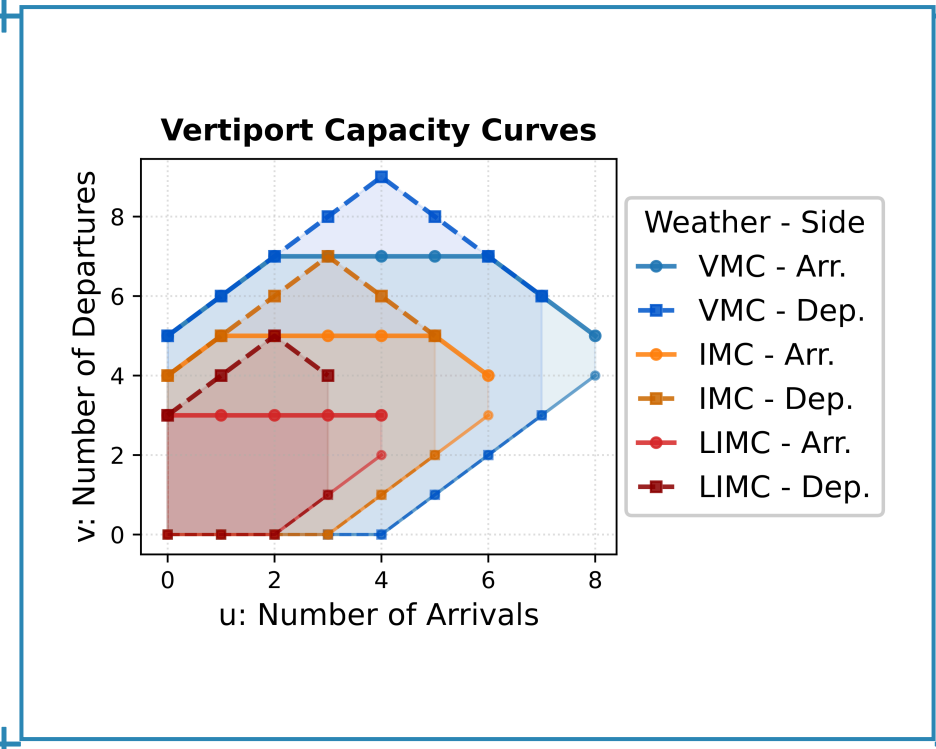
$$\mathbf{wc}_t \sim P(\mathbf{wc}_t \mid \mathbf{wc}_{t-1}, \hat{\mathbf{wc}}_t) \propto P(\mathbf{wc}_t \mid \mathbf{wc}_{t-1}) \cdot P(\mathbf{wc}_t \mid \hat{\mathbf{wc}}_t)$$

The wind state is sampled the same way, after converting wind speed and direction into crosswind and tailwind components relative to runway orientation.

Capacity modeling



SBGR runway envelopes, fitted by quantile regression (Ramanujam & Balakrishnan, 2009): concave and invertible — zero departures is always feasible.



Vertiport curves. Limited parking and aircraft availability impose a **minimum** departure count per arrival level, so the envelope is non-invertible.

TRAINING & EVALUATION

Train on forecast, test on observation

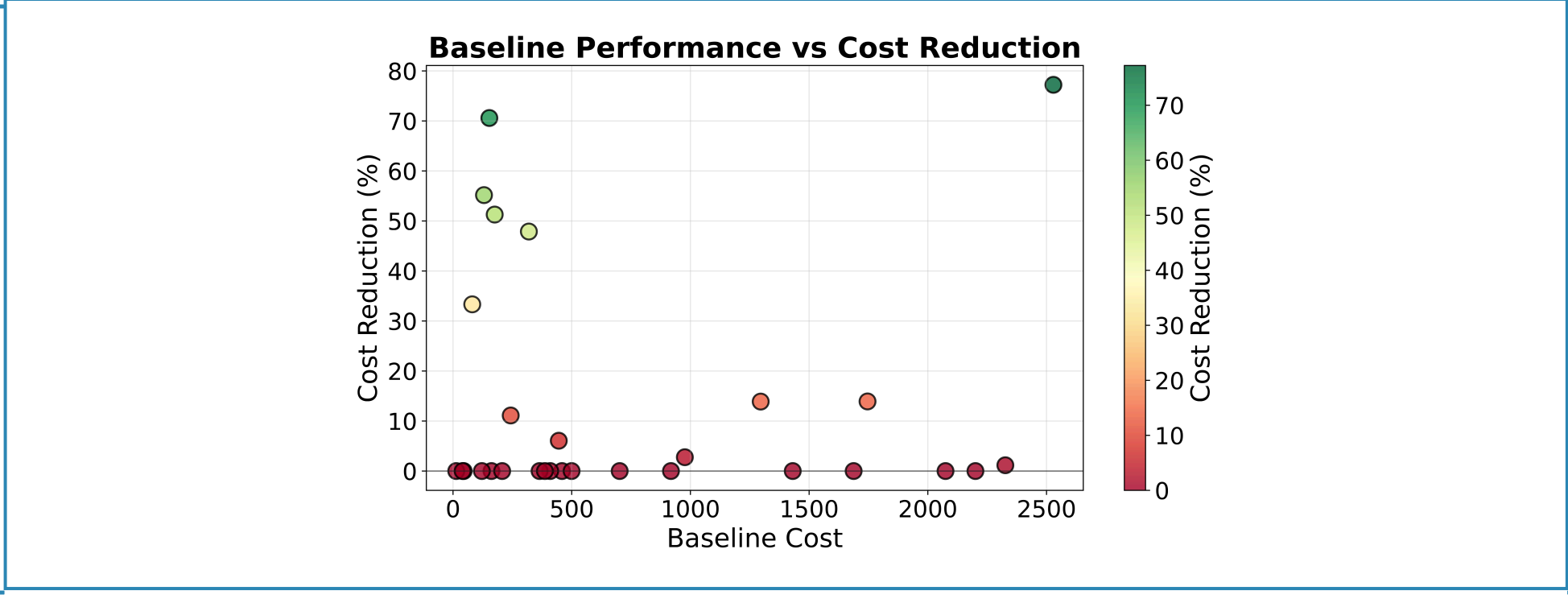
ALGORITHM Maskable PPO Stable Baselines 3 · Gymnasium	EPISODES PER DAY 500,000 conventional 300,000 UAM
DATA 24 months of METAR/TAF (REDEMETS) at 15-min resolution; ANAC SIROS schedules corrected with VRA and FlightRadar24	TEST SET 30 randomly selected days with diverse weather and demand; TAF for training, METAR for testing

Baseline — no rescheduling: the same tactical heuristic and capacity model with $g_f = 0$ for every flight. The strategic rescheduling cost is omitted in evaluation, so the metric isolates tactical congestion reduction.

RESULTS

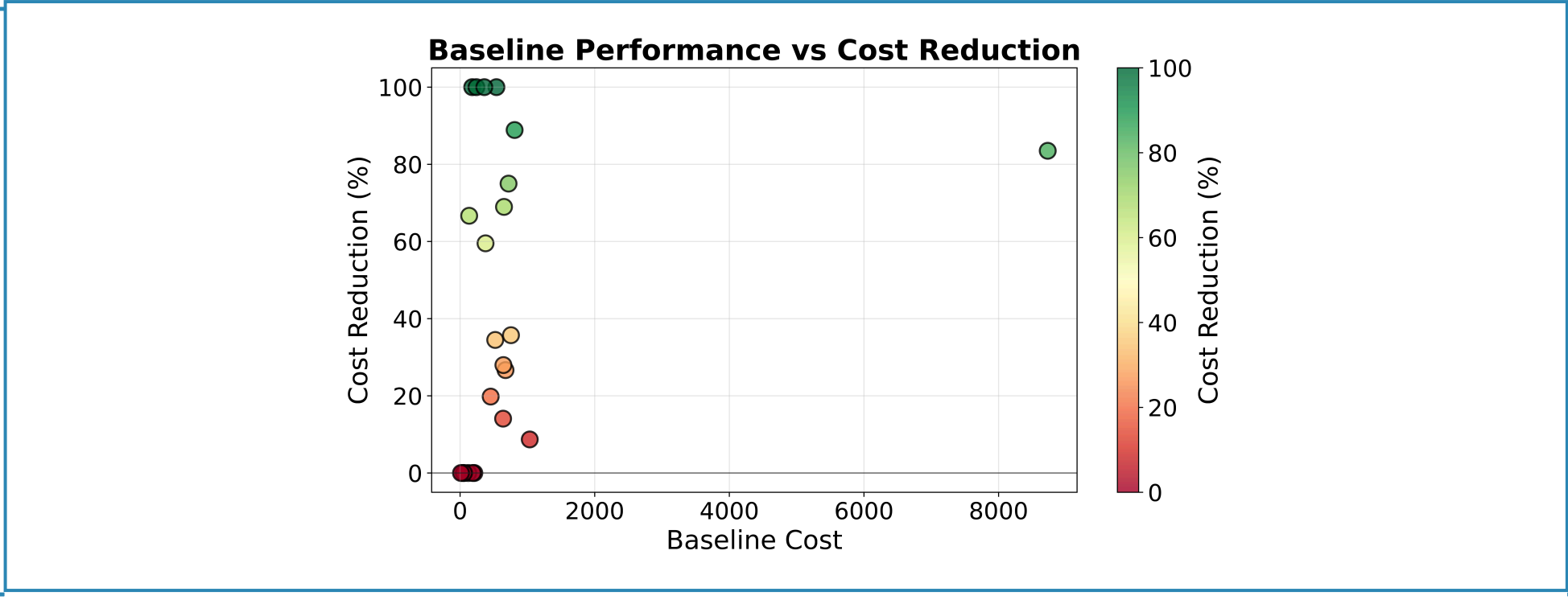
30 held-out days per case study

CASE 1 — CONVENTIONAL ATM AT SBGR		
13.3% aggregate mean-cost reduction	12/30 days improved	77.2% best single day
	0 days degraded	1.1% of flights delayed



The policy matches the baseline on most days and delivers large gains on a minority, including ~77% on the most congested day. Wilcoxon signed-rank $\mathbf{p} = \mathbf{0.002183}$. Of 10,144 flights, 98.9% fly as scheduled — a conservative policy that spends delay only where it removes real tactical congestion.

CASE 2 — UAM TRAFFIC MANAGEMENT, RMSP		
64.7% aggregate mean-cost reduction	23/30 days improved	43.9% mean daily reduction
	83.5% worst-case cost cut	4.4% of flights delayed



Widespread reductions, several complete eliminations of the daily tactical cost, and ~83% on the most congested day; maximum daily cost falls from €8,730 to €1,440. Wilcoxon signed-rank $\mathbf{p} = \mathbf{0.000281}$. The synthetic UAM baseline is deliberately not pre-optimized against vertiport capacity, leaving more headroom than an operationally planned schedule.

CONTRIBUTION 01

A model-free RL framework for strategic DCB that handles weather uncertainty through probabilistic weather representation and empirically grounded demand scenarios.

CONTRIBUTION 02

Integrated strategic and tactical optimization: ground delay allocation coupled with configuration and service-rate decisions.

CONTRIBUTION 03

Action masking that encodes valid ground-delay actions, accelerating learning and removing infeasible exploration.

CONTRIBUTION 04

Demonstrated transfer across conventional ATM and a UAM case study with the same algorithm and no retuning.

GROUND DELAY ALLOCATED — 10,144 / 2,857 FLIGHTS

DELAY	SBGR	UAM
0 slots	98.9%	95.6%
1 slot (15 min)	0.7%	2.2%
2 slots (30 min)	0.4%	1.8%
3 slots (45 min)	0.0%	0.2%
4 slots (60 min)	0.0%	0.3%

FUTURE WORK — BEYOND A SINGLE AIRPORT

Network effects, airline recovery, passenger connections and multi-airport coordination remain unexplored, as do wind-state definitions at other aerodromes.

FUTURE WORK — WEATHER REPRESENTATION

Route-level low-altitude weather and finer variables would improve realism over VMC/IMC/LIMC categories derived from airport METAR and TAF.